Exercise 3.2

SOLUTION

نوٹ

پہلے دی گئی equations کو Standard form میں لائیں پھر حل کریں

Q1. Solve the following equations.

i. $\frac{1}{3x}+\frac{4x}{6}=1\ ,\ x\neq 0$

Sol: $\frac{1}{3x}+\frac{4x}{6}=1\Rightarrow\frac{2+4x^2}{6x}=1$

$2+4x^2=6x\Rightarrow 4x^2-6x+2=0$

$4x^2-4x-2x+2=0\Rightarrow 4x(x-1)-2(x-1)=0$

$(4x-2)(x-1)=0\Rightarrow x-1=0\ \ or\ 4x-2=0$

$x=1\ \ or\ 4x=2\Rightarrow x=\frac{\cancel{2}}{\cancel{4}}=\frac{1}{2}$

$S.S=\left\{1,\frac{1}{2}\right\}$

ii. $\frac{x}{x+1}+\frac{x+1}{x}=\frac{5}{2};x\neq -1,0$

Sol: $\frac{x^2+(x+1)^2}{x(x+1)}=\frac{5}{2}$

$\frac{x^2+x^2+2x+1}{x^2+x}=\frac{5}{2}$

$\frac{2x^2+2x+1}{x^2+x}=\frac{5}{2}$ (By cross multiplication)

$\Rightarrow\ 2(2x^2+2x+1)=5(x^2+x)$

$4x^2+4x+2=5x^2+5x$

$5x^2+5x-4x^2-4x-2=0$

$x^2+x-2=0$

$\Rightarrow\ x^2+2x-x-2=0$

$x(x+2)-1(x+2)=0$

$\Rightarrow\ (x+2)(x-1)=0$

$x+2=0\ \ or\ x-1=0$

$x=-2\ \ or\ x=1$

$S.S=\{-2,1\}$

iii. $\frac{1}{x+1}+\frac{2}{x+2}=\frac{7}{x+5};x\neq -1,-2,-5$

Sol: $\frac{1(x+2)+2(x+1)}{(x+1)(x+2)}=\frac{7}{x+5}$

$\frac{x+2+2x+2}{x^2+3x+2}=\frac{7}{x+5}$

$\frac{3x+4}{x^2+3x+2}=\frac{7}{x+5}$

$\Rightarrow\ (3x+4)(x+5)=7(x^2+3x+2))$

$3x^2+15x+4x+20=7x^2+21x+14$

$7x^2+21x+14-3x^2-19x-20=0$

$4x^2+2x-6=0$

$4x^2+6x-4x-6=0$

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$2x(2x+3)-2(2x+3)=0$

$(2x+3)(2x-2)=0$

$2x+3=0 \ or \ 2x-2=0$

$x=\frac{-3}{2} \ or \ x=1$

$S.S=\left\{\frac{-3}{2},1\right\}$

iv. $\frac{a}{ax-1}+\frac{b}{bx-1}=a+b; x\neq\frac{1}{a},\frac{1}{b}$

Sol: $or \quad \frac{a}{(ax-1)}-b+\frac{b}{bx-1}-a=0$

$\frac{a-b(ax-1)}{(ax-1)}+\frac{b-a(bx-1)}{bx-1}=0$

$\frac{a-abx+b}{(ax-1)}+\frac{b-abx+a}{bx-1}=0$

$(a-abx+b)\left[\frac{1}{ax-1}+\frac{1}{bx-1}\right]=0$

$(a-abx+b)\left(\frac{bx-1+ax-1}{(ax-1)(bx-1)}\right)=0$

$(a-abx+b)(ax+bx-2)=0$

$a-abx+b=0 \ \ or \ \ ax+bx-2=0$

$abx=a+b \quad or \quad x(a+b)=2$

$x=\frac{a+b}{ab} \quad or \quad x=\frac{2}{a+b}$

$S.S=\left\{\frac{a+b}{ab},\frac{2}{a+b}\right\}$

v. $\frac{x+1}{x-1}+\frac{x-1}{x+1}=2 \ , \ x\neq1 \ , \ x\neq-1$

Sol: $\frac{x+1}{x-1}+\frac{x-1}{x+1}-2\Rightarrow\frac{(x+1)^2+(x-1)^2}{(x-1)(x+1)}=2$

$\frac{x^2+2x+1+x^2-2x+1}{x^2-1}=2\Rightarrow x^2+1-x^2+1=0$

$\Rightarrow 2=0$ not possible so, S.S = { }

vi. $3x^2+15x-2\sqrt{x^2+5x+1}=2$

Sol: $3\left(x^2+5x\right)-2\sqrt{x^2+5x+1}=2$

Put $\sqrt{x^2+5x+1}=y\Rightarrow x^2+5x+1=y^2$

$\Rightarrow x^2+5x=y^2-1$

We get $3\left(y^2-1\right)-2y=2\Rightarrow 3y^2-3-2y-2=0$

$3y^2-2y-5=0\Rightarrow 3y^2-5y+3y-5=0$

$y(3y-5)+1(3y-5)=0\Rightarrow(3y-5)(y+1)=0$

$3y-5=0 \ \ or \ \ y+1\Rightarrow y=\frac{5}{3} \ \ or \ \ y=-1$

When $y=\frac{5}{3}$ then $\sqrt{x^2+5x+1}=\frac{5}{3}$

$\Rightarrow x^2+5x+1=\frac{25}{9}\Rightarrow 9x^2+45x+9=25$

$9x^2+45x-16=0$

$x=\frac{-45\pm\sqrt{(45)^2-4(9)(-16)}}{2(9)}$

$x=\frac{-45\pm\sqrt{2025+576}}{18}=\frac{-45\pm\sqrt{2601}}{18}=\frac{-45\pm51}{18}$

$x=\frac{-45+51}{18} \quad and \quad x=\frac{-45-51}{18}$

$x=\frac{6}{18}=\frac{1}{3} \quad and \quad x=\frac{-96}{18}=\frac{-16}{3}$

When $y=-1$ then $\sqrt{x^2+5x+1}=-1$

$\Rightarrow x^2+5x+1=1\Rightarrow x^2+5x+1-1=0$

$x(x+5)=0\Rightarrow x=0 \ \ or \ x+5=0$

$x=0 \ \ or \ x=-5$

Checking

$3x^2+15x-2\sqrt{x^2+5x+1}=2$

For $x=0$

$3(0)^2+15(0)-2\sqrt{(0)^2+5(0)+1}=2$

$-2=2$ False

For $x=-5$

$3(-5)^2+15(-5)-2\sqrt{(-5)^2+5(-5)+1}=2$

$75-75-2\sqrt{25-25+1}=2\Rightarrow -2=2$ False

For $x=\frac{1}{3}$

$3\left(\frac{1}{3}\right)^2+15\left(\frac{1}{3}\right)-2\sqrt{\left(\frac{1}{3}\right)^2+5\left(\frac{1}{3}\right)+1}=2$

$3\left(\frac{1}{9}\right)+15\left(\frac{1}{3}\right)-2\sqrt{\left(\frac{1}{9}\right)+\frac{5}{3}+1}=2$

$\frac{1}{3}+5-2\sqrt{\frac{1+15+9}{9}}=2$

$\frac{1+15}{3}-2\sqrt{\frac{25}{9}}=2\Rightarrow\frac{16}{3}-2\left(\frac{5}{3}\right)=2$

$\frac{16-10}{3}=2\Rightarrow\frac{6}{3}=2\Rightarrow 2=2$ True

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For $x=-\frac{16}{3}$

$$3\left(\frac{-16}{3}\right)^2+15\left(\frac{-16}{3}\right)-2\sqrt{\left(\frac{-16}{3}\right)^2+5\left(\frac{-16}{3}\right)+1}=2$$

$$3\left(\frac{256}{9}\right)+15\left(\frac{-16}{3}\right)-2\sqrt{\frac{256}{9}-\frac{80}{3}+1}=2$$

$$\frac{256-240}{3}-2\sqrt{\frac{256-240}{9}+1}=2$$

$$\frac{16}{3}-2\sqrt{\frac{256-240+9}{9}}=2\Rightarrow\frac{16}{3}-2\sqrt{\frac{25}{9}}=2$$

$$\frac{16}{3}-2\left(\frac{5}{3}\right)=2\Rightarrow\frac{16}{3}-\frac{10}{3}=2$$

$$\Rightarrow\frac{6}{3}=2\Rightarrow 2=2$$

S.S $\left\{\frac{1}{3},\frac{-16}{3}\right\}$ and Extraneous roots 0 and -5

vii. $\sqrt{2x+8}+\sqrt{x+5}=7$

Sol: $\sqrt{2x+8}+\sqrt{x+5}=7$ _____ I

Squaring both sides

$$2x+8+x+5+2\sqrt{2x+8}\sqrt{x+5}=49$$

$$3x+13+2\sqrt{(2x+8)(x+5)}=49$$

$$2\sqrt{2x^2+10x+8x+40)}=49-13-3x$$

$$2\sqrt{2x^2+18x+40}=36-3x$$

Again Squaring

$$4(2x^2+18x+40)=1296+9x^2-216x$$

$$8x^2+72x+160=9x^2-216x+1296$$

$$9x^2-216x+1296-8x^2-72x-160=0$$

$$x^2-288x+1136=0$$

$$x^2-4x-284x+1136=0$$

$$x(x-4)-284(x-4)=0$$

$$(x-4)(x-284)=0$$

$$x-4=0 \quad or \quad x-284=0$$

$$x=4 \quad or \quad x=284$$

CHECKING for x = 4, (I) become

$$\sqrt{2(4)+8}+\sqrt{4+5}=7$$

$$\sqrt{16}+\sqrt{9}=7 \Rightarrow 4+3=7$$

$$\Rightarrow 7=7 \ TRUE$$

For x=284 (I) become

$$\sqrt{2(284)+8}+\sqrt{284+5}=7$$

$$\sqrt{256+8}+\sqrt{289}=7$$

$$\sqrt{276}+\sqrt{289}=7$$

$$\sqrt{276}+\sqrt{289}=7$$

24+17 = 7 **FALSE**

$S.S=\{4\}$ *and* Extraneous Root = 284

> نوٹ
>
> جس equation میں جذر ہو اُسے radical equation کہتے ہیں
>
> اور اس کی checking لازمی ہوتی ہے

viii. $\sqrt{3x+4}=2+\sqrt{2x-4}$

Sol: $\sqrt{3x+4}=2+\sqrt{2x-4}$

Squaring both sides

$$3x+4=4+2x-4+2(2)\sqrt{2x-4}$$

$$3x+4-2x=4\sqrt{2x-4}$$

$$x+4=4\sqrt{2x-4}$$

Again Squaring

$$x^2+8x+16=16(2x-4)$$

$$x^2+8x+16=32x-64$$

$$x^2+8x+16-32x+64=0$$

$$x^2-24x+80=0$$

$$x^2-4x-20x+80=0$$

$$x(x-4)-20(x-4)=0$$

$$(x-4)(x-20)=0$$

$$x-4=0 \quad or \quad x-20=0$$

$$x=4 \quad or \quad x=20$$

Checking for x=4 (I) become

$$\sqrt{3(4)+4}=2+\sqrt{2(4)-4}$$

$$\sqrt{16}=2+\sqrt{4}$$

$$4=2+2=4 \Rightarrow 4=4 \ True$$

For x=20 (I) become

$$\sqrt{3(20)+4}=2+\sqrt{2(20)-4}$$

$$\sqrt{60+4}=2+\sqrt{40-4}$$

$$\sqrt{64}=2+\sqrt{36} \Rightarrow 8=2+6\Rightarrow 8=8$$ **TRUE**

$S.S=\{4,20\}$

x. $\sqrt{x+7}+\sqrt{x+2}=\sqrt{6x+13}$

Sol: $\sqrt{x+7}+\sqrt{x+2}=\sqrt{6x+13}$ _____ I

Squaring both sides

$$x+7+x+2+2\sqrt{x+7}\sqrt{x+2}=6x+13$$

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$2x+9+2\sqrt{(x+7)(x+2)}=6x+13$

$2\sqrt{x^2+7x+2x+14}=6x+13-2x-9$

$\not{2}\sqrt{x^2+9x+14}=4x+4=\not{2}(2x+2)$

$\sqrt{x^2+9x+14}=2x+2$

Again Squaring

$x^2+9x+14=4x^2+4+8x$

$4x^2+8x+4-x^2-9x-14=0$

$3x^2-x-10=0$

$3x^2-6x+5x-10=0$

$3x(x-2)+5(x-2)=0$

$(x-2)(3x+5)=0$

$x-2=0 \quad or \quad 3x+5=0$

$x=2 \quad or \quad x=-5/3$

CHECKING for x = 2 I become

$\sqrt{2+7}+\sqrt{2+2}=\sqrt{6(2)+13}$

$\sqrt{9}+\sqrt{4}=\sqrt{25} \Rightarrow 3+2=5$

5 = 5 **TRUE**

For x = -5/3 I become

$\sqrt{\frac{-5}{3}+7}+\sqrt{\frac{-5}{3}+2}=\sqrt{6\left(\frac{-5}{3}\right)+13}$

$\sqrt{\frac{-5+21}{3}}+\sqrt{\frac{-5+6}{3}}=\sqrt{\frac{-30}{3}+13}$

$\sqrt{\frac{16}{3}}+\sqrt{\frac{1}{3}}=\sqrt{\frac{-30+39}{3}}$

$\frac{4}{\sqrt{3}}+\frac{1}{\sqrt{3}}=\sqrt{\frac{9}{3}} \Rightarrow \frac{5}{\sqrt{3}}=\sqrt{3}$ **FALSE**

$S.S=\{2\}$ *and* Extraneous Root $=\frac{-5}{3}$

x. $\sqrt{x+5}-\sqrt{x-3}=2$

Sol: $\sqrt{x+5}-\sqrt{x-3}=2$

Squaring both sides

$\left(\sqrt{x+5}-\sqrt{x-3}\right)^2=(2)^2$

$\left(\sqrt{x+5}\right)^2+\left(\sqrt{x-3}\right)^2-2\sqrt{x+5}\sqrt{x-3}=4$

$x+5+x-3-2\sqrt{(x+5)(x-3)}=4$

$2x+2-2\sqrt{x^2-3x+5x-15}=4$

$\not{2}\left(x+1-\sqrt{x^2+2x-15}\right)=2\cdot\not{2}$

$x+1-2=\sqrt{x^2+2x-15}$

$x-1=\sqrt{x^2+2x-15}$

Squaring again

$(x-1)^2=\left(\sqrt{x^2+2x-15}\right)^2$

$x^2-2x+1=x^2+2x-15$

$\not{x^2}+2x-15-\not{x^2}+2x-1=0 \Rightarrow 4x-16=0$

$4x=16 \Rightarrow x=4$

Checking $\sqrt{x+5}-\sqrt{x-3}=2$

Put

$x=4 \Rightarrow \sqrt{4+5}-\sqrt{4-3}=2 \Rightarrow \sqrt{9}-\sqrt{1}=2$

$3-1=2 \Rightarrow 2=2$ true

$S.S=\{4\}$

نوٹ

جو چیز معلوم کرنی ہے اسے x فرض کریں تو عبارت

سے equation بن جائے گی۔

Q2. **A former bought some sheep for Rs.9000. If he had paid Rs.100 less for each, he would have got 3 sheep more for the same money. How many sheep did he buy, when the rate in each case is uniform?**

Sol: Let number of sheep = x

Rate of x *sheep* $=9000$

Rate of one *sheep* $=\frac{9000}{x}$

If three sheep are more,

then rate of one sheep $=\frac{9000}{x+3}$

According to the given condition equation is

$\frac{9000}{x}-100=\frac{9000}{x+3} \Rightarrow \frac{9000-100x}{x}=\frac{9000}{x+3}$

$\Rightarrow (x+3)(9000-100x)=9000x$

$\Rightarrow 9000x-100x^2+27000-300x=9000x$

$\Rightarrow \not{9000x}-\not{9000x}+100x^2-27000+300x=0$

$\Rightarrow 100x^2+300x-27000=0$

$x^2+3x-270=0 \quad (\div)\ by\ 100$

$\Rightarrow x^2+18x-15x-270=0$

$\Rightarrow x(x+18)-15(x+18)=0$

$\Rightarrow (x+18)(x-15)=0$

$\Rightarrow x=-18\ or\ x=15$

-18 *Not possible so* $x=15=$ no of sheep's

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Q3. **A man sold his stock of eggs for Rs. 2400. If he had 2 dozen more, he would have got the same money by selling the whole for Rs. 0.50 per dozen cheaper. How many dozen eggs did he sell?**

Sol: Suppose man sold x dozen eggs.

Rate of x dozen =2400

Rate of one dozen $= \frac{2400}{x}$

Rate of one dozen if sold 2 dozen more for some many $= \frac{2400}{x+2}$

According to the given condition

$$\frac{2400}{x} - 0.5 = \frac{2400}{x+2} \Rightarrow \frac{2400}{x} - \frac{1}{2} = \frac{2400}{x+2}$$

$$\Rightarrow \frac{4800-x}{2x} = \frac{2400}{x+2}$$

$$\Rightarrow (x+2)(4800-x) = 2400(2x)$$

$$\Rightarrow 4800x - x^2 + 9600 - 2x = 4800x$$

$$x^2 + 2x + \cancel{4800x} - \cancel{4800x} - 9600$$

$$\Rightarrow x^2 + 2x - 9600 = 0$$

$$= \frac{-2 \pm \sqrt{(2)^2 - 4(1)(-9600)}}{2(1)} = \frac{-2 \pm \sqrt{4 + 38400}}{2}$$

$$x = \frac{-2 \pm \sqrt{38404}}{2}$$

$$x = \frac{-2 \pm 196(about)}{2}$$

$$\Rightarrow \frac{196-2}{2} \quad and\ x = \frac{-196-2}{2}$$

(Ignore -ve) not possible

$$x = \frac{194}{2} = 97 \quad and\ x = \frac{-198}{2} = -99$$

He sold 97 dozen eggs.

Q4. **A cyclist travelled 48km at a uniform speed. If he had travelled 2 km/hour slower, he would have taken 2 hours more to perform the journey. How long did he take to cover 48km?**

Sol: Let speed = v ; Time = t

According to the given condition equation is:

$Distance = S = vt = 48 — I \ \&\ (v-2)(t+2) = 48 — II$

$(two\ km\ slow = v-2\ two\ hour\ more = t+2)$

$II \Rightarrow vt + 2v - 2t - 4 = 48$

$(put\ vt = 48)\ \cancel{48} + 2v - 2t - 4 - \cancel{48} = 0$

$2v - 2t - 4 = 0 \quad ((\div)\ by\ 2)$

$v - t - 2 = 0 \Rightarrow v = t + 2 \text{——} III$

$Put\ III\ in\ I \Rightarrow vt = 48 \Rightarrow (t+2)t = 48$

$t^2 + 2t - 48 = 0$

$t^2 + 8t - 6t - 48 = 0$

$t(t+8) - 6(t+8) = 0$

$(t+8)(t-6) = 0$

$(ignore, t = -8) \Rightarrow \boxed{t = 6}$

Q5. **To do a piece of work, Abdullah takes 10 days more than Abdul Hadi. Together they finish the work in 12 days. How long would Abdul Hadi take to finish it alone?**

Sol: $Let\ Abdul\ Hadi\ finishes\ in = x\ days.$

$Let\ Abdullah\ finishes\ in = x+10$

$$Hadi's\ one\ days\ work = \frac{1}{x}$$

$$Abdullah's\ one\ days\ work = \frac{1}{x+10}$$

Abdullah's one days work $= \frac{1}{x+10} + \frac{1}{x}$

According to the given conditions equation is

$$\frac{1}{x+10} + \frac{1}{x} = \frac{1}{12}$$

$$\Rightarrow \frac{x+x+10}{x(x+10)} = \frac{1}{12} \quad \Rightarrow \quad \frac{2x+10}{x(x+10)} = \frac{1}{12}$$

(By cross multiplication) $12(2x+10) = x(x+10)$

$$\Rightarrow 24x + 120 = x^2 + 10x$$

$$\Rightarrow x^2 + 10x - 24x - 120 = 0$$

$$\Rightarrow x^2 - 14x + 120 = 0$$

$$\Rightarrow x^2 - 20x + 6x - 120 = 0$$

$$\Rightarrow x(x-20) + 6(x-20) = 0$$

$$\Rightarrow (x-20)(x+6) = 0$$

$x - 20 = 0 \ \ or \ \ x + 6 = 0$

$\Rightarrow x = 20 \ \ or \ \ x = -6$ (Not Possible)

$\Rightarrow x = 20$ Abdul Hadi finish work

Q6. **The braking distance (in metres) of a car is modeled by: $d(s) = 0.02s^2 + 0.1s$, where s is the speed of car in km/h. if the maximum safe braking distance is 50 metres, find the range of speed where braking is safe.**

Sol: According to the given conditions

$d(s) \le 50 \Rightarrow 0.02s^2 + 0.1s \le 50$

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$s^2 + 5s - 2500 \le 0 \left(\div by\ 0.02\right)$

To find values of s

$$\Rightarrow s = \frac{-5 \pm \sqrt{(5)^2 - 4(1)(-2500)}}{2(1)}$$

$$s = \frac{-5 \pm \sqrt{25 + 10000}}{2}$$

$$s = \frac{-5 \pm \sqrt{10025}}{2} \Rightarrow s = \frac{-5 \pm 100.12}{2}$$

$(ignore\ due\ to + ve)$

$$s = 47.56 < 0 \quad and \quad s = \frac{-115.12}{2}$$

$(0 \le s \le 47.560)\ km/h$

Q7. **A rocket follows the height function $h(t) = -5t^2 + 20t + 30$, where h(t) is the height in metres and t is the time in seconds. Find the time interval during which the rocket is at least 40 metres above the ground.**

Sol: According to the given conditions

$h(t) \ge 40 \Rightarrow -5t^2 + 20t + 30 \ge 40$

$\Rightarrow -5t^2 + 20t + 30 - 40 \ge 0$

$\Rightarrow -5t^2 + 20t - 10 \ge 0$

$\Rightarrow 5t^2 - 20t + 10 \le 0 \left('\times' by -1\right)$

$\Rightarrow t^2 - 4t + 2 \le 0\ \left(\div by\ 5\right)$

here a=1 , b=-4 , c=2

$$t = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(2)}}{2(1)} \le 0$$

$$\Rightarrow t = \frac{4 \pm \sqrt{16 - 8}}{2(1)} \le 0$$

$$t = \frac{4 \pm \sqrt{8}}{2} \le 0 \Rightarrow t = \frac{4 \pm 2.828}{2} \le 0$$

$$t = \frac{4 + 2.828}{2} \le 0 \quad and \quad t = \frac{4 - 2.828}{2} \le 0$$

$$t = \frac{6.828}{2} \le 0 \quad and \quad t = \frac{1.172}{2} \le 0$$

$t = 3.414 \le 0 \quad and \quad t = 0.586 \le 0$

$[0.586\ \text{sec} \quad , \quad 3.414\ \text{sec}]$

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